

ELSENBORN-BUTGENBACH AP, Belgium

WMO: 064960

Lat: 50.4822N

Lon: 6.1814E

Elev: 1870

StdP: 13.73

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	16.8	6.2	8.1	16.6	11.6	10.6	19.9	24.7	42.4	22.2	39.8	4.1	70	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	81.1	64.4	77.2	62.7	73.7	61.6	66.2	76.9	64.5	74.1	62.8	71.3	6.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	91.2	70.0	61.0	86.0	67.7	59.4	81.2	66.0	31.9	76.9	30.5	73.9	29.3	71.5	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.8	16.6	14.7	DB	3.2	87.1	6.2	3.0	-1.3	89.2	-5.0	91.0	-8.5	92.7	-13.0	94.8
			WB	2.6	70.8	6.2	2.1	-1.8	72.3	-5.4	73.5	-8.9	74.7	-13.4	76.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	32.3	33.0	37.9	43.8	50.8	55.8	59.5	58.5	52.7	47.0	39.5	34.1
	DBStd	11.66	7.49	7.67	6.77	6.87	6.49	6.18	5.82	5.64	5.39	6.48	6.66	7.16
	HDD50	2677	550	476	377	208	70	13	1	3	30	136	320	493
	HDD65	7171	1015	896	841	635	442	282	191	216	371	558	766	958
	CDD50	1026	0	0	1	23	94	188	296	267	110	43	4	0
	CDD65	43	0	0	0	0	1	7	20	14	1	0	0	0
	CDH74	720	0	0	0	3	30	124	281	255	26	1	0	0
	CDH80	155	0	0	0	0	3	20	66	65	1	0	0	0
	WSAvg	6.7	8.1	7.8	7.5	6.7	6.6	6.0	5.6	5.2	5.7	6.6	7.1	7.6
	PrecAvg	41.3	4.2	3.5	3.1	2.6	3.0	3.2	3.5	3.6	3.2	3.1	3.7	4.6
(15) Precipitation	PrecMax	53.1	10.1	10.2	6.6	6.7	6.6	6.4	8.3	8.1	9.0	9.0	6.1	11.9
	PrecMin	30.2	0.2	0.6	0.4	0.0	1.1	0.9	1.3	1.1	0.4	1.1	0.2	0.6
	PrecStd	5.9	2.1	2.0	1.9	1.4	1.2	1.4	1.5	1.9	2.0	1.6	1.5	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	54.4	62.7	72.8	78.1	83.1	86.7	86.4	77.8	69.3	58.4	50.8
		MCWB	47.6	44.0	48.4	55.1	60.9	64.5	65.5	66.5	62.3	57.8	53.1	48.1
	2%	DB	46.8	48.5	57.7	67.3	71.8	77.6	81.2	80.5	72.5	63.9	53.5	48.6
		MCWB	45.2	43.5	46.8	53.1	57.7	62.9	64.7	64.5	60.7	56.5	50.3	46.8
	5%	DB	44.4	45.6	52.7	61.9	68.3	73.3	76.9	75.8	67.8	59.8	50.9	46.4
		MCWB	43.0	43.1	45.4	50.7	56.0	61.3	62.9	63.0	58.6	54.7	48.7	45.2
	10%	DB	41.8	43.3	48.7	57.4	64.3	69.1	73.1	71.4	63.3	56.6	48.7	43.9
		MCWB	40.4	41.4	44.3	48.6	54.5	59.3	62.0	60.7	56.8	53.3	47.0	42.7
	0.4%	WB	48.5	47.4	51.0	56.9	63.6	68.1	68.4	68.7	64.3	60.2	53.5	49.4
		MCDB	49.1	50.0	57.2	68.4	72.7	78.5	79.9	80.6	73.3	66.7	55.8	50.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.6	45.0	48.7	54.3	60.2	64.7	66.3	65.9	62.0	57.5	50.9	47.1
		MCDB	46.5	47.2	54.1	64.4	69.4	73.7	77.1	76.8	70.0	61.7	52.9	48.1
	5%	WB	43.1	43.3	46.7	51.7	57.4	62.6	64.5	63.8	59.9	55.6	49.1	45.3
		MCDB	44.1	45.2	51.1	59.3	65.4	70.9	74.3	73.2	66.4	59.0	50.7	46.4
	10%	WB	40.6	41.3	44.9	49.6	55.2	60.3	62.8	62.0	57.7	53.8	47.1	42.9
		MCDB	41.6	43.2	48.4	56.2	62.4	67.4	71.4	69.5	62.4	56.4	48.6	43.8
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	10.5	14.1	17.8	18.2	18.4	18.3	19.1	16.8	13.2	8.9	8.2
		MCDBR	10.0	15.2	24.0	27.5	26.7	27.4	27.7	28.5	25.6	19.4	13.0	9.8
		MCWBR	9.0	10.7	14.5	15.1	14.5	14.5	13.3	14.2	15.0	12.3	9.8	8.4
	5% WB	MCDBR	8.7	11.3	18.3	23.8	22.9	23.7	23.8	24.6	22.6	16.3	11.0	8.3
		MCWBR	8.4	9.1	12.3	13.7	13.1	13.4	12.3	13.5	13.7	10.9	9.1	7.8
(40) Clear-Sky Solar Irradiance	taub		0.294	0.316	0.347	0.379	0.387	0.389	0.389	0.383	0.356	0.336	0.312	0.291
	taud		2.455	2.402	2.352	2.294	2.306	2.313	2.304	2.347	2.410	2.467	2.468	2.438
	Ebn at Noon		239	261	271	273	276	275	273	269	265	250	228	221
	Edh at Noon		19	25	32	38	39	39	39	36	30	24	18	16
(44) All-Sky Solar Radiation	RadAvg		243	461	822	1307	1540	1692	1603	1374	1026	601	281	188
	RadStd		37	89	123	153	149	160	189	140	123	58	36	28

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page